

- [41] Kai Guo and Markus J Buehler. Rapid prediction of protein natural frequencies using graph neural networks. *Digital Discovery*, 1(3):277–285, 2022.
- [42] Alireza Ghafarollahi and Markus J. Buehler. Sciagents: Automating scientific discovery through multi-agent intelligent graph reasoning. 2024.
- [43] Qizhi Pei, Lijun Wu, Kaiyuan Gao, Jinhua Zhu, Yue Wang, Zun Wang, Tao Qin, and Rui Yan. Leveraging biomolecule and natural language through multi-modal learning: A survey. *arXiv preprint arXiv:2403.01528*, 2024.
- [44] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V Chawla, Olaf Wiest, and Xiangliang Zhang. Large language model based multi-agents: A survey of progress and challenges. *arXiv preprint arXiv:2402.01680*, 2024.
- [45] Junlin Xie, Zhihong Chen, Ruifei Zhang, Xiang Wan, and Guanbin Li. Large multimodal agents: A survey. *arXiv preprint arXiv:2402.15116*, 2024.
- [46] Yuheng Cheng, Ceyao Zhang, Zhengwen Zhang, Xiangrui Meng, Sirui Hong, Wenhao Li, Zihao Wang, Zekai Wang, Feng Yin, Junhua Zhao, et al. Exploring large language model based intelligent agents: Definitions, methods, and prospects. *arXiv preprint arXiv:2401.03428*, 2024.
- [47] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, pages 1–11, 2024.
- [48] Markus J Buehler. Generative retrieval-augmented ontologic graph and multiagent strategies for interpretive large language model-based materials design. *ACS Engineering Au*, 4(2):241–277, 2024.
- [49] Bo Ni and Markus J Buehler. Mechagents: Large language model multi-agent collaborations can solve mechanics problems, generate new data, and integrate knowledge. *Extreme Mechanics Letters*, 67:102131, 2024.
- [50] Isabella Stewart and Markus Buehler. Molecular analysis and design using multimodal generative artificial intelligence via multi-agent modeling. 2024.
- [51] Alireza Ghafarollahi and Markus J Buehler. Protagents: protein discovery via large language model multi-agent collaborations combining physics and machine learning. *Digital Discovery*, 2024.
- [52] Alireza Ghafarollahi and Markus J Buehler. Atomagents: Alloy design and discovery through physics-aware multi-modal multi-agent artificial intelligence. *arXiv preprint arXiv:2407.10022*, 2024.
- [53] Gabriele Corso, Luca Cavalleri, Dominique Beaini, Pietro Liò, and Petar Veličković. Principal neighbourhood aggregation for graph nets. *Advances in Neural Information Processing Systems*, 33:13260–13271, 2020.
- [54] Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2016.
- [55] Petar Velićković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, Yoshua Bengio, et al. Graph attention networks. *stat*, 1050(20):10–48550, 2017.
- [56] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? *arXiv preprint arXiv:1810.00826*, 2018.
- [57] Markus J. Buehler. Accelerating scientific discovery with generative knowledge extraction, graph-based representation, and multimodal intelligent graph reasoning. *Machine Learning: Science and Technology*, 2024.
- [58] Steve Plimpton. Fast parallel algorithms for short-range molecular dynamics. *Journal of computational physics*, 117(1):1–19, 1995.
- [59] Sheng Yin, Yunxing Zuo, Anas Abu-Odeh, Hui Zheng, Xiang-Guo Li, Jun Ding, Shyue Ping Ong, Mark Asta, and Robert O Ritchie. Atomistic simulations of dislocation mobility in refractory high-entropy alloys and the effect of chemical short-range order. *Nature communications*, 12(1):4873, 2021.
- [60] Emile Maras, Oleg Trushin, Alexander Stukowski, Tapio Ala-Nissila, and Hannes Jonsson. Global transition path search for dislocation formation in ge on si (001). *Computer Physics Communications*, 205:13–21, 2016.
- [61] Aiichiro Nakano. A space–time-ensemble parallel nudged elastic band algorithm for molecular kinetics simulation. *Computer Physics Communications*, 178(4):280–289, 2008.